

# Methodology, Estimation & Replication Notes

Technical companion to “Firm- and Regime-Conditioned NLP Alpha”

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This appendix supports the 2-page proposal. It is an external research agenda inspired by challenges in modern text-based systematic investing. Performance figures are reference results from prior published literature (gross unless stated) and are **not** a backtest by the author; “diagnostics” below are *planned*, clearly tagged **OURS**, not completed results.

## A. Literature Review & Positioning

The supervised approach of **Ke, Kelly & Xiu (SESTM, 2019)** learns a sentiment scoring rule from text using realised returns as weak labels: a marginal screen selects sentiment-charged words, a two-topic model represents positive- and negative-sentiment language, and a penalised likelihood scores each article. It is transparent and, on Dow Jones Newswires, reports a large equal-weight long–short Sharpe with low factor  $R^2$  **PRIOR-LIT**. **Loughran–McDonald** supplies a fixed finance dictionary; **Tetlock** and **Jegadeesh–Wu** established media tone as return-relevant; **Gentzkow–Kelly–Taddy** survey text-as-data; **Gu–Kelly–Xiu** place text alongside other predictors in ML asset pricing. More recent work fine-tunes **LLM text representations** for return prediction and reports that aggregated decoder-LLM embeddings can outperform conventional sentiment scores **PRIOR-LIT**.

### Where these methods may leave value on the table (the basis for this proposal)

- They fit a *market-wide* language→sentiment map; firm context is largely discarded (§B).
- They treat polarity as *time-stationary* within an estimation window; regime dependence is not modelled (§C).
- They output a *ranking*; calibrated magnitude and uncertainty are absent (§D).
- LLM-based variants are sensitive to **look-ahead and survivorship bias**: under contaminated or non-point-in-time evaluation, reported LLM-based alpha can be materially overstated (§K). This argues for content-structural, interpretable improvements rather than opaque scaling.

The agenda below is motivated by limitations the author has encountered when replicating these published methods on point-in-time data. Through extensive replication of this literature and discussion with researchers working on financial NLP, the author has developed a detailed understanding of where these methods succeed and where they encounter practical limitations. The framing is deliberately academic: these are research hypotheses with falsifiable diagnostics, not claims about any specific firm’s production systems.

## B. Methodology I — Firm-Conditioned Sentiment, at the Topic/Event Level **OURS**

**Premise.** A word’s return implication is firm-dependent: “restructuring” may signal a credible turnaround for a distressed value name and value destruction for a leveraged growth name. A market-wide map books the cross-sectional average and is least accurate precisely in the idiosyncratic names where news matters most.

**Why not a raw firm×word table.** With a vocabulary of  $10^4$ – $10^5$  terms and thousands of firms, a free firm-word sentiment matrix is hopelessly sparse and would overfit. The estimable formulation operates at the level of  $K$  *topic/event clusters* (phrase clusters formed from co-occurrence or embedding similarity, e.g. *guidance-cut*, *buyback*, *litigation*, *refinancing*), with  $K \approx 200$ – $500 \ll$  vocabulary size). The object estimated is the firm-specific *deviation* from a market-wide cluster dictionary  $\{s_c\}$  — low-rank and shrunk — not a free, separate dictionary per firm; this is what makes it estimable from pooled data.

**Model.** Let  $s_{i,c}$  be the sentiment of cluster  $c$  for firm  $i$ :

$$s_{i,c} = s_c + e_i^\top W h_c, \quad W = UV^\top, \quad U, V \in \mathbb{R}^{d \times r}, \quad r \ll d,$$

where  $s_c$  is the market-wide cluster sentiment,  $e_i \in \mathbb{R}^d$  is a point-in-time *firm characteristic embedding* (sector, size, book-to-market, leverage, profitability, momentum, lifecycle), and  $h_c \in \mathbb{R}^d$  is a fixed *cluster embedding* from a text encoder. The article score aggregates cluster loadings,

$$p_i = \sigma\left(\sum_c a_{i,c} s_{\text{firm}(i),c}\right), \quad a_{i,c} \geq 0, \quad \sum_c a_{i,c} = 1,$$

where  $a_{i,c}$  is the share of article  $i$ ’s screened content assigned to cluster  $c$ .

## What is estimated, and how

- **Parameters.**  $\{s_c\}_{c=1}^K$  (market-wide,  $K$  scalars) and the low-rank  $W = UV^\top$  ( $2dr$  parameters). With  $d \approx 16\text{--}32$ ,  $r \approx 2\text{--}5$ , this is  $O(K + 2dr)$  — hundreds to low thousands of parameters, versus  $O(K \times N_{\text{firms}})$  for an unpooled table. Firm and cluster embeddings  $e_i, h_c$  are treated as fixed throughout estimation.
- **Training sample.** Article-level rows  $(i, \text{firm}(i), t_i)$  over the training window, each weakly labelled by the standardised cross-sectional *rank* of the firm’s forward return (robust to heavy tails).
- **Objective.**  $\min_{\{s_c\}, U, V} \sum_i \ell(\text{rank}(r_i), p_i) + \lambda \|W\|_F^2 + \eta \|s\|_2^2$ , with  $\ell$  the calibrated loss of §G.4. The Frobenius penalty (equivalently a prior  $\delta_{i,c} \sim \mathcal{N}(0, \tau^2)$ ) shrinks the firm tilt toward the market map.
- **Regularisation choice.**  $\lambda$  (hence  $\tau$ ),  $r$ , and  $K$  are selected on a validation fold by empirical Bayes / cross-validation, not on test.
- **Estimation.** Alternating minimisation (convex in  $\{s_c\}$  given  $W$ ; smooth in  $U, V$ ) or joint SGD over mini-batches; scales to  $\sim 10\text{M}$  articles.

**Identification.** Center the firm tilt across firms,  $\sum_i \omega_i e_i = 0$ , so  $\mathbb{E}[\delta_{i,c}] = 0$  and  $s_c$  is identified as the market-wide level while  $e_i^\top W h_c$  is a mean-zero firm deviation. Low rank  $r$  and characteristic-based  $e_i$  prevent the deviation from absorbing idiosyncratic noise;  $W$  is identified up to rotation, resolved by an orthonormal  $U$ . A sector  $\times$  size cluster fallback covers firms with thin history.

**Computational feasibility.** Clusters and embeddings are precomputed once; the trainable parameter count is small; the per-epoch cost is linear in the number of (article, cluster) loadings. A first implementation is computationally tractable on standard research hardware.

**Planned diagnostics** OURS. (a) firm-conditioning ablation ( $\delta \equiv 0$  vs full) on out-of-sample IC; (b) decomposition of gains by size/sector to test the small-cap/idiosyncratic prediction; (c) inspection of the largest tilts  $e_i^\top W h_c$  for economic plausibility.

## C. Methodology II — Regime-Conditioned Language OURS

**Premise.** The *return relevance* of language is regime-dependent: balance-sheet and margin language (“cost reduction”, “debt paydown”) tends to be rewarded in downturns/tightening and discounted in expansions, and reactions to guidance amplify in high volatility. A static map can lag and degrade performance around regime transitions.

**State.** Define an *ex-ante* macro state  $z_t \in \mathbb{R}^m$  ( $m$  small,  $\approx 4\text{--}6$ ), observable at  $t$ : realised/implied volatility level and percentile, term spread, credit spread, trailing market return, and a tightening/easing indicator from policy-rate changes. All components are standardised on trailing data and strictly as-of  $t$ .

**Parameterisation.** Add a cluster-level regime term — *not* a per-word coefficient vector —

$$s_{i,c,t} = \underbrace{s_c}_{\text{market}} + \underbrace{e_i^\top W h_c}_{\text{firm tilt}} + \underbrace{\beta_c^\top z_t}_{\text{regime tilt}}, \quad \beta_c \in \mathbb{R}^m.$$

This adds only  $K \times m$  parameters. A firm  $\times$  regime interaction  $e_i^\top A(z_t) h_c$  is deliberately deferred to later work to keep the first model identifiable.

## What is estimated, and how

- **Parameters.**  $\{\beta_c\}$  ( $K \times m$ ), estimated jointly with §B or in a second stage, under a ridge penalty  $v \|\beta\|^2$  and partial pooling  $\beta_c \sim \mathcal{N}(0, \Sigma_\beta)$  across clusters.
- **Regime definition.** Kept *continuous* in  $z_t$ ; no hidden-Markov latent state is estimated (which would be fragile on few, persistent regimes). A continuous state uses all observations rather than partitioning a short sample.
- **Estimation & selection.** Penalised regression;  $v$  chosen on a validation block using **block / embargoed** time-series cross-validation that respects regime persistence (no peeking past  $t$ ).

**Identification.** Few effective regimes  $\Rightarrow$  small effective sample, so  $m$  is kept small,  $z_t$  standardised, and  $\beta_c$  shrunk hard and partially pooled. Additivity keeps the regime tilt separable from the firm tilt.

**Planned diagnostics** OURS. Regime-stratified IC and turnover; static vs additive-regime out-of-sample on regime-balanced folds; behaviour through 2008-/2020-type transitions.

## D. Methodology III — From Ranking to Calibrated, Sized Signals OURS

**Premise.** A cross-sectional rank answers *who* to favour; downstream sizing needs *how much* — a calibrated magnitude with uncertainty. This is a change of *objective*, not a portfolio model.

**Calibration map.** Learn a monotone map  $c(\cdot)$  from raw score to expected forward return on the training fold by **isotonic regression** or **Platt scaling**, fit over score buckets, giving  $\hat{\mu}_i = c(p_i)$  comparable across names and dates. Quality is checked with a **reliability (calibration) curve**: bucketed predicted vs realised returns.

**Heavy tails  $\Rightarrow$  robust objectives.** Equity returns are heavy-tailed, so a plain Gaussian likelihood is a poor default. The scorer/sizer is fit under one of, and compared across:

- **Huber loss** — a robust conditional mean, down-weighting outliers;
- **Quantile (pinball) loss** — predictive intervals (e.g. 10/50/90), no symmetry assumption;
- **Student- $t$  negative log-likelihood** — a heavy-tailed location–scale fit;
- with selection by validation calibration and rank IC, not by in-sample fit.

**Confidence.** Build  $\hat{\sigma}_i$  (or full quantiles) from interpretable components: score extremity; corroboration (number/agreement of same-day articles); novelty (fresh vs echoed news); ensemble disagreement; and text informativeness (specific, quantitative language vs boilerplate).

**Evaluation (out-of-sample).** Reliability-curve fit; interval *coverage*; Spearman IC; **out-of-sample decile returns and realised hit rates**; decile monotonicity; signal decay.

**Delivery.** The output is a calibrated expected-return *proxy* and a reliability/conviction score —  $(\hat{\mu}_i, \hat{\sigma}_i)$  or a set of predictive quantiles, carried with the event-specific half-life of §E. This is deliberately *not* a claim of precise expected returns or position sizes: equity text is noisy, so a monotone score $\rightarrow$ return map and the signal’s reliability are estimated and validated out-of-sample, and *position sizing is left to downstream optimisation. The agenda stops at the calibrated signal*; sizing, risk, and execution remain with downstream teams (§F).

## E. Methodology IV — Information Decay Modelling OURS

**Premise.** A news signal is not equally useful over a fixed horizon: an earnings surprise may be largely priced within hours, whereas a regulatory investigation or a guidance change can inform for days or weeks. Applying one holding horizon to every signal mis-times entries, inflates turnover, and overstates capacity. The decay profile is itself *event-type-* and *regime-dependent*, and it is estimable.

**Model (parsimonious baseline specification).** Write the expected signal contribution at holding horizon  $h$  (trading days) as strength  $\times$  decay,

$$a_i(h) = \underbrace{g(p_i)}_{\text{calibrated strength (\$D)}} \cdot \underbrace{D_{e(i)}(h; z_t)}_{\text{decay kernel}}, \quad D_e(h) = e^{-h/\tau_e},$$

where  $e(i)$  is the article’s event cluster (the §B taxonomy) and  $\tau_e > 0$  is an event-type *half-life*. The exponential kernel is the baseline; a **piecewise-linear** kernel is the robust alternative; more flexible decay specifications are left for future work.

### What is estimated, and how

- **Observable input / target.** Event-tagged articles with point-in-time forward returns at horizons  $h \in \{1, 3, 5, 10, 21\}$  days; the target is the realised forward return (equivalently the signal’s horizon- $h$  information coefficient).
- **Parameters.** One half-life  $\tau_e$  per event cluster ( $K$  scalars); an optional small regime modifier  $\tau_e(z_t) = \tau_e e^{\gamma_e^\top z_t}$  *only* where an event type has enough per-regime observations.
- **Estimation.** Fit the horizon profile of the signal’s predictive coefficient / IC by non-linear least squares (exponential) or isotonic/piecewise regression on the training window; **pool** event types hierarchically and **shrink**  $\tau_e$  toward a global half-life  $\tau_0$ .
- **Small-sample behaviour.** Rare event types fall back to  $\tau_0$ ; regime-conditioning of  $\tau$  is added only when supported, otherwise the kernel stays regime-flat.
- **Feasibility & timing.** A handful of parameters, closed-form curve-fit, trivially implementable in Python (scipy curve-fitting / per-horizon regressions). Decay is estimated on *completed past* events in the training window and applied forward; horizon returns enter only the fit, never the live signal at  $t$  — no look-ahead.

**Production relevance.** The signal ships with a half-life, so delivery becomes  $(\hat{\mu}_i, \hat{\sigma}_i, \hat{\tau}_e)$ : downstream can match holding period to decay, blend horizons, and control turnover instead of assuming one fixed horizon. **Planned diagnostics** OURS : per-event IC( $h$ ) curves; out-of-sample stability of  $\tau_e$ ; turnover/capacity as a function of holding horizon.

## F. Methodology V — Production Pipeline, Monitoring & Retraining OURS

A point-in-time, modular workflow of the kind used in industrial-scale signal generation; every stage is reproducible and leakage-screened.

| Stage                      | Responsibilities / controls                                                                          | Main failure mode                                  |
|----------------------------|------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| <b>Data ingestion</b>      | point-in-time capture; exact timestamps; source/version provenance.                                  | missed or late items; silently revised timestamps. |
| <b>Text processing</b>     | entity resolution to a permanent id; dedup and <i>novelty</i> scoring; normalisation.                | mis-mapped entity; duplicate inflation.            |
| <b>Feature extraction</b>  | screened terms; firm-/regime-conditioned cluster features; optional LLM embeddings.                  | vocabulary drift; embedding-version skew.          |
| <b>Signal construction</b> | firm- & regime-conditioned scorer; calibration to $(\hat{\mu}_i, \hat{\sigma}_i)$ ; decay half-life. | over-fit firm tilt; mis-calibration.               |
| <b>Validation</b>          | walk-forward, regime-stratified; factor-orthogonality QC; turnover/decay; look-ahead screens.        | leakage; optimistic in-sample fit.                 |
| <b>Deployment</b>          | versioned signal-delivery contract; reproducible as-of snapshots.                                    | train/serve skew; stale features.                  |
| <b>Monitoring</b>          | input-distribution and IC/calibration drift; coverage; turnover alarms.                              | label latency; silent concept drift.               |
| <b>Retraining</b>          | scheduled rolling re-fit + alarm-triggered refresh; champion/challenger.                             | over-frequent refit; regime overfit.               |

Required inputs across stages: timestamped text, a point-in-time entity map and characteristic panel, multi-horizon returns, a factor panel for orthogonality QC, and logged features/signals for monitoring.

### Data sources — what each contributes

| Source                           | Latency / horizon | Information contributed                                                       |
|----------------------------------|-------------------|-------------------------------------------------------------------------------|
| <b>Dow Jones Newswire</b>        | low / short       | breaking, machine-readable events; first-mover signal.                        |
| <b>Bloomberg News</b>            | low / short       | broad market-moving headlines and commentary.                                 |
| <b>RavenPack</b>                 | low / short       | pre-tagged entities, events, novelty, relevance — a baseline to beat.         |
| <b>Earnings calls (FactSet)</b>  | scheduled / med   | management tone, Q&A, guidance — richest for firm-specific language (§B).     |
| <b>SEC filings (10-K/Q, 8-K)</b> | higher / long     | formal disclosure, risk factors, MD&A — precise, slow, complementary horizon. |
| <b>Alternative text</b>          | varies            | breadth (supply-chain, social) — noisier, non-linear information.             |

The firm-specific layer (§B) is strongest on transcripts and filings; breaking-news sources drive latency-sensitive signals; combining horizons is where incremental, orthogonal information is most likely to be found. A baseline implementation requires no proprietary feed beyond a point-in-time characteristic panel and a timestamped news/returns sample.

### Monitoring & concept drift OURS

**Premise.** Language and market structure evolve, so a static scorer degrades silently (concept drift); production NLP must monitor and refresh.

- **Monitor.** Input/feature-distribution drift (population-stability index, KS tests on embedding summaries), vocabulary-novelty rate, rolling IC / hit-rate, calibration-curve drift, coverage, and turnover. Input-distribution drift is observable *immediately*; return-based metrics lag by the label horizon.
- **Detect.** Thresholded alarms on rolling metrics; drift tests; a champion/challenger comparison against the incumbent model.
- **Retrain.** Scheduled rolling re-fit plus alarm-triggered retraining, always walk-forward (never on future data). *Failure mode:* label latency delays return-based detection, so input-drift signals bridge the gap.

## G. Estimation Details & Derivations

### G.1 SESTM foundation (prior literature) PRIOR-LIT

**Screen.** For word  $w$ , with the sign of the firm's labelled return defining “positive” articles,

$$f_w = \frac{\#\{\text{positive-return articles containing } w\}}{\#\{\text{articles containing } w\}}.$$

Retain words with  $|f_w - \frac{1}{2}|$  large and document frequency  $\geq k$ ; the rest are treated as noise.

**Two-topic generative model.** For an article with screened term counts  $d_i$  and length  $|d_i|$ ,

$$d_i \sim \text{Multinomial}(|d_i|, p_i O_+ + (1 - p_i) O_-),$$

with word distributions  $O_+, O_-$  over the screened vocabulary and article sentiment  $p_i \in [0, 1]$ .

**Penalised article scoring.**

$$\hat{p}_i = \arg \max_{p \in [0, 1]} \sum_w d_{i,w} \log(p O_{+,w} + (1 - p) O_{-,w}) + \lambda \log(p(1 - p)),$$

the penalty pulling scores toward  $\frac{1}{2}$  to suppress noise from short, low-information articles.

### G.2 Firm-conditioned extension OURS

With cluster sentiments  $s_{i,c} = s_c + e_i^\top W h_c$ ,  $W = UV^\top$ , fit  $(\{s_c\}, U, V)$  by weak-supervised regression on return ranks,

$$\min_{\{s_c\}, U, V} \sum_i \ell(\text{rank}(r_i), p_i) + \lambda \|W\|_F^2, \quad p_i = \sigma\left(\sum_c a_{i,c} s_{\text{firm}(i),c}\right).$$

### G.3 Regime-conditioned extension OURS

The additive state term gives  $s_{i,c,t} = s_c + e_i^\top W h_c + \beta_c^\top z_t$ ; estimate  $\{\beta_c\}$  with ridge  $v\|\beta\|^2$  and partial pooling  $\beta_c \sim \mathcal{N}(0, \Sigma_\beta)$ , with  $z_t$  using only information available at  $t$ .

### G.4 Calibrated, distributional output OURS

Calibrate  $\hat{\mu}_i = c(p_i)$  with isotonic/Platt  $c$ . Fit the magnitude/uncertainty under a robust objective rather than a Gaussian default — e.g. the quantile (pinball) loss for level  $\tau$ ,

$$\rho_\tau(u) = u(\tau - \mathbb{I}\{u < 0\}), \quad u = r_i - \hat{q}_{i,\tau},$$

estimated at several  $\tau$  for a predictive interval, or a Huber / Student- $t$  likelihood; deliver  $(\hat{\mu}_i, \hat{\sigma}_i)$  or the quantile set  $\{\hat{q}_{i,\tau}\}$ .

### G.5 Information-decay kernel OURS

With event-cluster half-lives,  $D_e(h) = e^{-h/\tau_e}$ ; estimate  $\{\tau_e\}$  by non-linear least squares on the horizon- $h$  IC profile (or a piecewise-linear fit), pooled across event types and shrunk toward a global  $\tau_0$ , adding a regime modifier  $\tau_e(z_t) = \tau_e e^{\gamma_e^\top z_t}$  only where per-regime data permit. The half-life is delivered alongside  $(\hat{\mu}_i, \hat{\sigma}_i)$  so the holding horizon can be matched to decay.

## H. Replication Notes & Observed Limitations

**What was replicated.** A transparent SESTM-style pipeline (screen  $\rightarrow$  two-topic  $\rightarrow$  penalised MLE), a Loughran–McDonald dictionary baseline, a commercial-feed-style sentiment baseline, and an embedding-based return baseline, under walk-forward, next-open execution. Qualitatively, the supervised approach reproduces intuitive, stable word lists and a strong equal-weight long–short with a much weaker value-weight leg and high turnover.

**Benchmark figures (reference results from prior literature, gross) PRIOR-LIT.** As reported by Ke, Kelly & Xiu (2019) on Dow Jones Newswires: an equal-weight long–short Sharpe on the order of 4.3 with daily average return exceeding 30 bps and factor  $R^2$  below roughly 10%; the supervised score above a commercial feed and a dictionary on the same news; and the strategy described as exploitable at reasonable turnover net of transaction costs once turnover is controlled. The sample is large (multi-decade newswire coverage; on the order of 10M single-firm-tagged articles). These figures are taken from the cited working paper's empirical results, are gross unless stated, and are **not** measured by the author; the exact magnitudes depend on the paper's universe and period and should be read against its tables.

**Limitations that motivate §B–D (observed in replication / practice) OURS.**

- Sign errors tend to cluster in firms/sectors where a word’s meaning is firm-dependent  $\Rightarrow$  §B.
- Performance can be unstable across regime shifts; rolling re-estimation responds with a lag  $\Rightarrow$  §C.
- Rank-only output does not answer the sizing questions downstream teams ask  $\Rightarrow$  §D.
- Much of the practical risk is timestamp/entity leakage and silent decay rather than modelling  $\Rightarrow$  §E, §F.

## I. Feasible First Implementation

The proposed agenda is organised as a sequence of incremental extensions. A realistic sequence, each phase gated on the previous:

| Phase                | Scope                                                                                                                                                  | Success criterion                                                             |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| 1 Baseline           | Replicate SESTM, Loughran–McDonald, a commercial-feed-style score, and an embedding baseline; strict point-in-time, walk-forward, next-open execution. | Reproduces stable word lists and a comparable OOS ranking edge.               |
| 2 Firm-conditioned   | Move from word to topic/event clusters; add the low-rank firm tilt $e_i^\top W h_c$ with characteristic shrinkage.                                     | Positive OOS IC uplift vs Phase 1, concentrated in small/idiosyncratic names. |
| 3 Regime-conditioned | Add the additive ex-ante term $\beta_c^\top z_t$ (continuous state; no HMM; parsimonious additive specification).                                      | Improved regime-stratified IC and stability through transitions.              |
| 4 Calibration        | Map scores to $(\hat{\mu}_i, \hat{\sigma}_i)$ via isotonic/Platt with a robust (Huber / quantile / Student- $t$ ) objective.                           | Reliable calibration curve and interval coverage.                             |
| 5 Validation         | Full signal report: IC, hit rate, decay, turnover, capacity proxy, small-cap concentration, sector neutrality, factor exposure; gross and net.         | Incremental, orthogonal, net-positive signal with documented exposures.       |

If an extension fails to add out-of-sample, net-of-cost value over the previous phase, that negative result is itself a deliverable and the extension is not shipped. Information-decay estimation (§E) attaches to Phases 4–5; blind-input validation (§K) runs alongside Phases 1–2 as a robustness check; concept-drift monitoring (§F) is a deployment-time addition rather than part of the first paper.

## J. Self-Critique — Risks & Mitigations OURS

The main risks and how the design addresses them:

| Risk                               | Mitigation                                                                                                                                                                   |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Identification (firm/regime tilts) | Center the firm tilt; low rank $r$ ; small $m$ ; shrink and partially pool; adopt an additive regime specification to preserve separability between firm and regime effects. |
| Overfitting via hyper-parameters   | All of $\lambda, v, r, K$ chosen on a validation block; headline results out-of-sample only; minimal grids.                                                                  |
| Sparse language estimation         | Operate at $K$ topic/event clusters, not raw words; characteristic-similarity pooling; sector $\times$ size fallback for thin firms.                                         |
| Regime instability / few regimes   | Continuous $z_t$ (no latent-state estimation); embargoed, regime-aware CV; hard shrinkage on $\beta_c$ .                                                                     |
| Computational burden               | Frozen embeddings, precomputed clusters, low-rank $W$ , mini-batch SGD; small trainable parameter count.                                                                     |
| Data requirements / leakage        | Point-in-time characteristic panel and timestamped news/returns; entity-resolution and timestamp screens before any label is attached.                                       |
| Heavy-tailed targets               | Robust losses (Huber / quantile / Student- $t$ ) and rank-based weak labels rather than a Gaussian default.                                                                  |
| Alpha-decay estimability           | Few parameters (one half-life per event cluster); exponential / piecewise baseline; hierarchical pooling toward a global half-life; neural decay deferred.                   |
| Concept drift / non-stationarity   | Input-distribution and rolling-IC monitoring; scheduled + alarm-triggered retraining; champion/challenger.                                                                   |
| Identity memorisation / leakage    | Blind / anonymised-input validation; entity masking; point-in-time and entity-resolution screens.                                                                            |

## K. Modern Extensions & Current-Literature Scan

Directions from current practice in production-oriented text-to-signal pipelines, kept where they fit the text→signal mandate.

- **LLM text representations as features** **FUTURE** . Aggregated decoder-LLM embeddings have been reported to outperform conventional sentiment scores for return prediction (Guo & Hauptmann, 2024); they could feed the firm/regime layer as inputs while the scoring stays interpretable, rather than treating an LLM as an end-to-end oracle.
- **Look-ahead bias as a binding constraint** **FUTURE** . Frontier LLMs are trained across long time spans, so naive backtests can be temporally contaminated, and reported gains may reflect memorisation, survivorship, or retrospective edits. Mitigations in the recent literature include point-in-time retrieval over archived, as-of-dated corpora with temporal-violation screens, exclusion of easily-recalled observations, smaller or inference-time *unlearning*-adjusted models, and evaluation by alpha-decay across regimes. Under such protocols, reported LLM-based alpha can be materially overstated (Glasserman & Lin, 2023; Merchant & Levy, 2025; FINSABER, 2025) — therefore treated as a primary modelling constraint.
- **Blind / anonymised inputs — a validation framework** **FUTURE** . To test whether alpha comes from textual *content* rather than company *memorisation*, score an entity-anonymised copy of the news — firm names, tickers, and people replaced by neutral placeholders (“Company A”) via NER — and measure the performance drop. A large drop indicates the model reads content; a small drop flags reliance on identity/ticker priors (a leakage and robustness red flag, acute for LLM components, and consistent with the “distraction effect” of Glasserman & Lin, 2023). This is a *validation diagnostic*, not a standalone signal; anonymisation is imperfect (context can re-identify), so it is read directionally, and it is implementable with standard NER plus known name/ticker lists.
- **Retrieval-augmented, point-in-time generation** **FUTURE** . RAG over filings/transcripts with reranking and long-context handling is useful for grounding, but only with strict as-of provenance (Lee & Roh, 2024).
- **Event-aware & multi-modal signals** **FUTURE** . Condition sentiment on event type (M&A, guidance, litigation) and fuse text with fundamentals/price context.
- **Agentic / multi-agent tooling** **FUTURE** . Multi-agent LLM systems are promising as *research accelerators* and as a mechanism for accelerating hypothesis generation and data collection under point-in-time constraints — gated by the same point-in-time validation, not used as an unchecked signal source.

## L. References

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